I'll mention here that tensor products and product states are two different things, despite their similar-sounding names.[2]

A tensor product is a *vector space* for studying composite systems. A product state is a *state-vector*. It's one of the many state-vectors that inhabit a product space. As we will see, most of the state-vectors in the product space are *not* product states.

6.6 COUNTING PARAMETERS FOR THE PRODUCT STATE

Let's consider the number of parameters it takes to specify such a product state. Each factor requires two complex numbers (α_u and α_d for Alice, β_u and β_d for Bob), which means we need four complex numbers altogether. That's equivalent to eight real parameters. But recall that the normalization conditions in Eqs. 6.4 reduce this by two. Furthermore, the overall phases of each state have no physical significance, so the total number of real parameters is four. That's hardly surprising: it took two parameters to describe the state of a single spin, so two independent spins require four.

6.7 ENTANGLED STATES

The principles of quantum mechanics allow us to superpose basis vectors in more general ways than just product states. The most general vector in the composite space of states is

$$\psi_{uu}|uu\rangle + \psi_{ud}|ud\rangle + \psi_{du}|du\rangle + \psi_{dd}|dd\rangle,$$

where we have used the subscripted symbols ψ (instead of α and β) to represent the complex coefficients. Again, we have four complex numbers, but this time we only have one normalization condition,

$$\psi^*_{uu}\psi_{uu} + \psi^*_{ud}\psi_{ud} + \psi^*_{du}\psi_{du} + \psi^*_{dd}\psi_{dd} = 1,$$